Course Title	Electronic Circuit Design	**Course Code**	EE2001
Department	Department of Electrical Engineering (DEE)	**Campus**	Lahore
Knowledge Profile	Engineering Fundamentals (WK3)	**Credit Hrs.**	3+1
Knowledge Area	Electronics (KA04)	**Grading Scheme**	Relative
HEC Knowledge Area	Breadth Cores	**Applicable From**	Spring 2023
SDG	4 | Quality Education	**PBL**	1
Pre-requisite(s)	EL1004, EE1004		

Course Objective	To build the fundamental concepts for analysis and design of electronic circuits, both analog and digital.

No.	Assigned Program Learning Outcome (PLO)
1	An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
3	An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

I = Introduction, R = Reinforcement, E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Analyze basic amplifier circuit models.	Q1, M1, F	C4	1
2	Design electronic circuits using op-amps.	A1, M1, F	C5	3
3	Appraise the limitations of real op-amp circuits	Q2, M2	C4	1
4	Design wave-form generation circuits.	Q3, A2, M2, F	C5	3
5	Design CMOS logic circuits.	Q4, F	C5	3

Text Books	**Title**	Microelectronic Circuits, Theory and Applications
	Author	Adel S. Sedra
	Publisher	Oxford University Press, 6th Edition
Reference Books	**Title**	Design with Operational Amplifiers and Analog ICs
	Author	Sergio Franco
	Publisher	McGraw Hill, 3rd Edition

Week	Course Contents/Topics	Chapter*	CLO*
01	Introduction, Amplifiers	1	1
02	Amplifiers Circuit Models, Frequency Response	1	1
03	Operational Amplifiers: The ideal Op-Amp, Inverting and Non-inverting Amp, Difference Amplifier	5	2
04	Effect of Finite Open-Loop Gain, Large signal Operation of Op-Amp	5	2
05	Non-Linear Operations, Op-Amp applications	5	2
06	Integrators and Differentiators	5	2
07	DC Imperfections in real op-amps (slew rate, offset voltage, bias current)	5	3
8	Effect of DC Imperfections on op-amp integrators	5	3
9	Signal and Waveform Generators, Wien-Bridge Oscillator	12	4
10	Bistable Multivibrator, Astable Multivibrator	12	4
11	Monostable Multivibrator, Function Generator	12	4
12	IC Timers and its applications	12	4
13	Precision Rectifiers	12	4
14	Digital circuit Design, CMOS Circuits	10	5
15	CMOS Logic Gates, Performance analysis	10	5
16	Latches and flip-flops, Random Access Memories, Emitter Coupled Logic	14	5

***Reference book chapters are given in brackets**

Assessment Tools	Weightage
Quizzes, Assignments	20.0%
Midterms (I+II)	30.0%
Final Exam	50.0%